

JAY MUTUM

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SUMMARY

I am a University of Minnesota (Twin Cities) Computer Science major graduate, and currently pursuing a master's degree in Computer Science at the University of Illinois Urbana-Champaign.

I am a full stack developer with multiple years of programming experience since my high school years, and looking for an entry level employment as a software developer and to expand my employment experience and knowledge of computer science. I have a good foundational and working knowledge of cloud computing. I have four professional Azure certifications. I am looking for development and leadership experience in the workforce as part of a team.

EDUCATION

Masters of Computer Science, UNIVERSITY OF ILLINOIS AT URBANA CHAMPAIGN May 2025-Current
Major: Computer Science GPA: 4 / 4.00

Bachelor of Science, UNIVERSITY OF MINNESOTA, TWIN CITIES COLLEGE OF SCIENCE AND ENGINEERING May 2024
Major: Computer Science GPA: 3.9 / 4.00 Dean's List: 2022, 2023, 2024
Recipient of Gold Scholar Award, National Scholarship, and National Merit Scholarship.

High School Diploma, LISLE HIGH SCHOOL, ILLINOIS May 2021
National Merit Scholar Finalist. Illinois State Scholar. Math Team Captain and Science Olympiad Member 2017-2021.
Officer, Student Ambassadors 2019-2021.

TECHNICAL SKILLS

- **BACK-END DEVELOPMENT** | Python, Java, C/C++
- **DATABASE** | PostgreSQL, MySQL, SQL Server
- Experience in Java
- Worked as part of a team for university coding projects
- Communication
- Azure Cloud certifications
- Leadership
- **FRONT-END DEVELOPMENT** | HTML, JavaScript, CSS, React
- Used Python to implement forms of AI (supervised and unsupervised)
- Used C++ in a team project
- Language: Spanish Seal of Biliteracy
- Agile/Scrum Methodology

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals
- Microsoft Certified: AI Fundamentals
- Microsoft Certified: Data Fundamentals
- Microsoft Certified: Security, Compliance, and Identity Fundamentals

PROJECTS

Alzheimer's Disease Classification Using Deep Learning & Clinical Domain Knowledge

- Reproduced and implemented a healthcare ML research pipeline for Alzheimer's disease classification using high-resolution 3D MRI data from the ADNI dataset, leveraging clinically informed brain-region selection and convolutional neural networks (CNNs).
- Developed and evaluated 3D CNN, Topological Data Analysis (TDA), and interpretability approaches using Python, TensorFlow/Keras, PyHealth, and Grad-CAM; performed ablation studies across full-brain, image-patch, and hippocampal representations.
- Demonstrated that focusing on the clinically relevant left hippocampus achieved competitive classification performance (~84% accuracy), while substantially reducing computational complexity compared with more complex architectures.

Applied Cryptography & Cybersecurity Attack Analysis

- Implemented Python-based cryptographic tools and attack techniques involving AES, RSA, MD5/SHA-1/SHA-256, and cryptographic hash functions using PyCryptoDome and related security libraries.
- Exploited real-world cryptographic weaknesses including weak-key brute forcing, hash collisions, MD5 length-extension attacks, AES padding-oracle attacks, and RSA vulnerabilities; generated colliding programs and certificates to demonstrate practical security impacts.
- Developed command-line Python scripts to automate encryption/decryption, collision detection, cryptanalysis, and attack workflows, strengthening practical understanding of secure authentication, cryptographic implementation pitfalls, and offensive security.

Simplification of Image Data (Matlab)

- Principal Component Analysis to simplify data.
- Recreated images with features accounting for differing amounts of variance.
- Generated graphics to show increased visual clarity as proportion of variance increased in order to identify the most essential features of the data.

Drone Delivery System Simulation (C++, Node.js)

- Implemented a drone delivery system simulation model to handle pathing algorithms.
- Application was deployed on client server architecture with requests simulated through a webpage.
- Tested the model with 5000 automated requests.

Data Analysis of Spam and Real Comments on YouTube (Pandas, Numpy Libraries, DataFrame)

- Developed modules to differentiate between spam and real comments on YouTube.
- Classified data using naive bayes, neural network and clustering and parsed data using pandas and numpy libraries.
- Classification methods all had >90% accuracy across popular YouTube music videos (Gangnam Style, Roar, etc.)

Website (HTML, CSS, JavaScript, Python, SQL)

- Developed a website using HTML, CSS, and JavaScript for the front-end.
- Implemented backend with node.js (express).
- Utilized PostgreSQL for database handling on the backend.
- Implemented features like user login and time entry.
- Optimized website performance, resulting in a 30% reduction in page load times.

Testing and Validating of User Input

- Simulated testing for validation and sanitization of user input.
- Used the GoogleTest Framework to automate test cases for data validation.
- Created testing documentation. Documented use cases and test results from test suite and retested results.

Reinforcement Learning with Python

- Developed algorithms within aim-python environment models.
- Developed value and policy iteration algorithms for the “gridworld” environment.
- Created and modified pathing algorithms for the traveling salesman problem in Python.

ADDITIONAL EXPERIENCE

JUNE – AUGUST 2022, NOVEMBER 2024 – PRESENT (PART-TIME)

E-COMMERCE SHOPPER, WHOLE FOODS

My duties as an e-commerce shopper included fulfilling Amazon Prime Now customer orders, preparing Whole Foods Market products for delivery and/or pickup, and other in-store duties as assigned.

JUNE – JULY 2018

SUMMER READING VOLUNTEER, LISLE LIBRARY

Helped with a summer reading program, which is aimed to get kids interested and having fun in reading. I used interpersonal skills when talking to both parents and kids alike, and registering them for the program. Handed out and tallied prizes for reading.